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Olkhov, Victor

Independent

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Victor Olkhov

Independent, Moscow, Russia

victor.olkhov@gmail.com

ORCID: 0000-0003-0944-5113

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Abstract

We show that Markowitz's (1952) portfolio variance describes only a limited case when all volumes of successive trades at exchange with shares of all securities of the portfolio are assumed constant during the averaging interval. We overcome this limitation and derive market-based portfolio variance that depends upon the means, variances, and covariances of random values and volumes of consecutive trades. To derive that, we convert the time series of trades made at the exchange with shares of portfolio securities and obtain the time series that model the trades with the portfolio like with a single security. That establishes an equal description of market-based variance of any security and portfolio. If all trade volumes are assumed constant, the market-based variance coincides with Markowitz' variance. We show the ties between the market-based variance of the portfolio and its Sharpe Ratio and describe the dependence of the portfolio Sharpe Ratio on random trade volumes. We prove that correlations between random prices and trade volumes always equal zero. To study market-based statistical dependence between random trade volumes and prices, one should empirically calculate correlations between prices and squares of trade volumes.

Keywords : portfolio variance, portfolio theory, random trade volumes

JEL: C0, E4, F3, G1, G12

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Introduction

The modern portfolio theory and optimal portfolio selection have their origin in the classical Markowitz (1952) paper. Since 1952, Markowitz's mean-variance portfolio selection frame was enriched with the dependence on the transaction costs and taxes (Pogue, 1970), higher moments (Samuelson, 1970), lifetime portfolio selection (Merton, 1969), and many others. Markowitz's Nobel Prize presentation (Markowitz, 1991) was complemented by 50- (Rubinstein, 2002) and 70-year reviews (Boyd et al., 2024).

The derivation of the portfolio variance $\Theta_M(t,t_0)$ (1) by Markowitz was a central piece of optimal portfolio selection that served as the assessment of the portfolio risks for 75 years.

$$\Theta_M(t,t_0) = \sum_{j,k=1}^{J} \theta_{jk}(t,t_0) X_j(t_0)\, X_k(t_0) \tag{1}$$

The coefficients $\theta_{jk}(t,t_0)$ (1) at the current time t denote the covariances of returns of securities j and k, and $X_j(t_0)$ denotes the relative amount invested into security j at time t_0 in the past.

In this paper we restudy the derivation of Markowitz's portfolio variance. However, we don't consider any developments of optimal portfolio selection or extensions of portfolio theory.

In his famous study, Markowitz didn't specify any possible implementations or examples of the random variables that model stochastic returns of the securities that compose the portfolio. "*This paper does not consider the difficult question of how investors do (or should) form their probability beliefs.*" (footnote, p. 81, Markowitz, 1952). However, we believe that investors and portfolio managers consider the time series of values and volumes of consecutive trades made at the exchange with the securities that compose the portfolio as random. These random time series of the values and volumes of market trades made at the exchange determine the means and variances of prices and returns.

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In this paper we study how the expression of Markowitz's (1952) portfolio variance $\Theta_M(t,t_0)$ (1) is consistent with the reality of random trades at the exchange. To derive that, we convert the time series of trades made at the exchange with the portfolio securities and obtain the time series that model the trade with the portfolio, like with a single security. That reveals an equal description of the variance of any single security and of the portfolio that is composed of $j=1,..J$ securities, and we derive market-based portfolio variance that accounts for random volumes of the consecutive trades. If one assumes that all trade volumes are constant, then market-based portfolio variance takes the form of Markowitz variance $\Theta_M(t,t_0)$ (1).

The randomness of the volumes of consecutive trades made at the exchange is an essential property of financial markets. The use of a limited case of constant trade volumes can't be the basis for the reliable forecasts of mean and variance, which, in real markets, are determined by random values and volumes of consecutive trades. That may be one of the causes of low predictability of returns in current studies, which implicitly use constant trade volumes approximation. In some sense, the use of the limited approximation of constant trade volumes to predict the mean and variance of return that are determined by random volumes of consecutive trades is like the use of constant steps while predicting random walks. The reliability of results will be low.

Ultimately, the methods for optimal portfolio selections that are based on the forecasts, which implicitly use the assumption of constant volumes of successive trades, may cause the selected portfolio to be rather far from optimal.

The description of market-based variances of securities and portfolios, which account for random volumes of consecutive trades with securities, is more complex. Actually, the time of simple assessments and easy investment decisions is over. Those who prefer simple investment solutions may follow familiar decisions but should be ready for excess losses.

We consider market-based Sharpe Ratios of a security or of the portfolio and derive their dependence on the coefficients of variation of random trade volumes. Finally, we show that the use of volume weighted average price that accounts for the random trade volumes results in zero price-volume correlations. To study the empirical statistical dependence of random trade volumes and prices, one should calculate the correlations between random prices and *squares* of trade volumes.

We propose three options to read the article. To read the Summary for Practitioners (Section 6) and then conclude: is it worth learning economic and math justifications? The second option is to read first the main body of the article that is almost free from formulas. Those who are interested mostly in math derivations may start reading the Appendices.

In Section 1, we consider Markowitz's equation (1.2) that describes the dependence of the portfolio random return R on random returns R_j of securities that compose the portfolio. In Section 2, we obtain the time series that model the trade with a portfolio like with a single security. In Section 3, we show how the consideration of random market trades as the origin of random return determines the equation (3.1) that describes the dependence of portfolio random return $R(t_i,t_0)$ on random returns $R_j(t_i,t_0)$ of securities and on random coefficients $x_j(t_i)$ (3.1), which were missed in Markowitz's equation (1.2). Section 4 describes market-based variance that accounts for random trade volumes. In Section 5 we discuss the market-based Sharpe Ratio. Summary for Practitioners in Section 6. Conclusion in Section 7. In App. A, we derive the time series that model the trades with a portfolio like with a single security. App. B models random returns, mean returns, and market-based variance. App. C presents the Taylor series of the market-based Sharpe Ratio by the coefficient of variation of random volumes. In App. D, we prove that price-volume correlations are always zero. To study statistical dependence between random trade volumes and prices, one should calculate market-based correlations between random prices and squares of trade volumes. All prices are adjusted to current time t.

1. Markowitz Variance

Throughout this article we consider an investor who at time t_0 in the past has collected his portfolio with shares of $j=1,..J$ securities (A.1; A.2). Since then, the investor doesn't trade his shares and keeps his portfolio unchanged. However, at current time t the investor is interested in the assessments of the current return and variance of his portfolio. To do that, the investor may observe the time series of values, volumes, and prices (A.4; A.5) of market trades with each security of his portfolio during a particular averaging interval Δ (A.3).

We believe that most investors and portfolio managers regularly perform such or similar monitoring of their portfolios and model portfolio mean and variance as it was developed almost 75 years ago in a classical paper by Markowitz (1952). Markowitz derived the dependence of the mean return $R(t,t_0)$ (1.1) and the variance $\Theta_M(t,t_0)$ (1) of the portfolio on mean returns $R_j(t,t_0)$ (A.7) and covariances $\theta_{jk}(t,t_0)$ (1) of its securities.

$$R(t,t_0) = \sum_{j=1}^{J} R_j(t,t_0)\, X_j(t_0) \tag{1.1}$$

The coefficients $X_j(t_0)$ in (1.1) denote relative amounts invested into security j at time t_0. The relations (1; 1.1) describe the mean-variance Markowitz's frame for the optimal portfolio selection and are probably the most used relations in portfolio theory and practical investments. To derive the expressions of mean $R(t,t_0)$ (1.1) and the variance $\Theta_M(t,t_0)$ (1) of portfolio return, Markowitz, without any justifications or explanations at all, proposed the equation (1.2) that define the dependence of random portfolio returns R and random returns R_j of the securities that compose the portfolio. We exactly reproduce Markowitz's (1952) notations: "The yield R on the portfolio as a whole is:

$$R = \sum_{j=1}^{J} R_j X_j \tag{1.2}$$

The R_j (and consequently R) are considered to be random variables." (Markowitz, 1952, p.81, the 3rd equation from the top). The equation (1.2) directly determines the expressions of the mean $R(t,t_0)$ (1.1) and the variance $\Theta_M(t,t_0)$ (1). However, as we mentioned above, Markowitz didn't consider any possible implementations of random returns.

Meanwhile, not much has changed since 1952, and market trade was then and remains now the main and only economic and financial process that collects and implements the action of all possible disturbing factors, such as expectations, shocks, economic and monetary policies, etc., into time series of performed random market trades. The investors and portfolio managers consider market trades made at the exchange during the averaging interval as the only process that determines the random returns on their investments. The time series of random values and volumes of trades that are made at the exchange during a particular averaging interval with securities of the portfolio present the only real existing source of the randomness of the returns. The study of random properties of the trades made at the exchange during a particular averaging interval models the current mean and variance of returns and allows practitioners to update investment strategies and perform the forecasts of the mean and variance of returns.

In this paper we study the time series of random values and volumes of market trades with securities of the portfolio. That helps us to verify Markowitz's (1952) derivation of the mean and variance of returns. We consider Markowitz equation (1.2) as the central point for the derivation of the mean $R(t,t_0)$ (1.1) and the variance $\Theta_M(t,t_0)$ (1). We use the time series of the trades with securities to express Markowitz's equation (1.2) via random returns $R_j(t_i,t_0)$ (A.7) of security j of the trade that was made at time t_i during the averaging interval Δ (A.3).

In Apps. A and B, we demonstrate that the main equation (A.9) that determines the dependence of portfolio random returns $R(t_i,t_0)$ at time t_i during the averaging interval Δ (A.3) on random returns $R_j(t_i,t_0)$ of its securities describes only a rather limited case when all volumes of the consecutive trades with portfolio securities are assumed constant during Δ. For convenience, we give here (1.3; A.9) the presentation of Markowitz's equation (1.2) in terms of random returns $R_j(t_i,t_0)$ of securities as a result of random trades at time t_i during the interval Δ (A.3):

$$R(t_i, t_0) = \sum_{j=1}^{J} R_j(t_i, t_0) X_j(t_0) \tag{1.3}$$

The mathematical expectation of equation (1.3) determines equation (1.1) on mean return $R(t,t_0)$, which is always correct. As well, the equation (1.3) also determines Markowitz variance $\Theta_M(t,t_0)$ (1). However, as we show in App. B, the correct consideration of the time series of random market trades with securities of the portfolio leads us to the conclusion that the equation (1.4) that determines the dependence of random portfolio returns on random returns of its securities has a more complex expression (B.3; 1.4). Equation (1.4) arises the dependence of random portfolio return $R(t_i,t_0)$ not only on random returns $R_j(t_i,t_0)$ of the securities but also on random relative volumes $x_j(t_i)$ (B.3) of consecutive trades, which was missed in Markowitz's equation (1.2; 1.3):

$$R(t_i, t_0) = \sum_{j=1}^{J} R_j(t_i, t_0) \cdot X_j(t_0) \cdot \frac{x_j(t_i)}{x_j(t_0)} \quad ; \quad x_j(t_i) = \frac{u_j(t_i)}{W(t_i)} \tag{1.4}$$

In App. B, we show that Markowitz's equation (1.2) and equation (1.3) are only limited cases when all volumes of consecutive trades with all securities of the portfolio are constant during Δ (A.3) of the general equation (1.4). That determines the limitation of applicability of Markowitz portfolio variance $\Theta_M(t,t_0)$ (1) only for the case of constant trade volumes.

However, modern markets reveal highly irregular or random changes in the trade volumes, and the use of Markowitz's variance $\Theta_M(t,t_0)$ (1) for such cases may result in excess risks and losses. We overcome the limitations of applicability of Markowitz's variance $\Theta_M(t,t_0)$ (1) to the case of constant trade volumes only and derive the market-based variances $\Theta(t,t_0)$ (B.18) of the securities and portfolios, which accounts for random volumes of consecutive trades. Moreover, we show that there is no difference between the assessments of the performance of a security or a portfolio composed of many securities.

2. Model of Trades with Portfolio as with a Single Security

To model the trades with the portfolio as with a single security, one should consider the trades per share of the portfolio. Let us consider the portfolio that had $U_{\Sigma j}(t_0)$ (A.1) shares of security j at time t_0 in the past. At current time t, the investor observes the trades with security j and assesses the total trade volume $U_{\Sigma j}(t)$ (A.5) during the averaging interval Δ (A.3). We highlight that one should always consider the averaging interval Δ (A.3), during which the total trade volume $U_{\Sigma j}(t)$ (A.5) with shares of security j is always much more than the number $U_{\Sigma j}(t_0)$ (A.1) of its shares in the portfolio. That requirement guarantees that the total number of shares of all securities $j=1,..J$ traded during Δ (A.3) will be much more than the number of all shares $W_\Sigma(t_0)$ (A.2) of investor's portfolio.

The investor may multiply each volume $U_j(t_i)$ of trades made during Δ by the ratio of the number $U_{\Sigma j}(t_0)$ (A.1) of its shares in the portfolio at time t_0 to the total trade volume $U_{\Sigma j}(t)$ (A.5) during Δ. The resulting time series of rescaled trade volumes $u_j(t_i)$ (A.10) at time t have the total trade volume $u_{\Sigma j}(t)$ (A.11) during Δ, which exactly equals the number $U_{\Sigma j}(t_0)$ (A.1) of shares of security j in the portfolio at time t_0. The investor may convert the time series of values $C_j(t_i)$ (A.4) of trades with the same security j by multiplying each trade value $C_j(t_i)$ by the same ratio of the number $U_{\Sigma j}(t_0)$ of shares to the total trade volume $U_{\Sigma j}(t)$ during Δ. The time series of rescaled values $c_j(t_i)$ and volumes $u_j(t_i)$ (A.10) of trades with a security j model the same prices $p_j(t_i)$ (A.4; A.10) as the initial time series of values $C_j(t_i)$ and volumes $U_j(t_i)$ trades and model the trade with exactly the number $U_{\Sigma j}(t_0)$ (A.1) of shares of security j in the portfolio.

The investor may convert the time series of values and volumes of trades with each security $j=1,..J$ of his portfolio in the same way. The prices $p_j(t_i)$ (A.10) of trades with securities don't change with the converting from trade values $C_j(t_i)$ and volumes $U_j(t_i)$ (A.4) to rescaled trade values $c_j(t_i)$ and volumes $u_j(t_i)$ (A.10). For each security j of the portfolio, the time series of rescaled values $c_j(t_i)$, volumes $u_j(t_i)$, and prices $p_j(t_i)$ (A.10) model market trades with the total volume $u_{\Sigma j}(t)$ (A.11) of trade during Δ, which exactly equals the number of shares $U_{\Sigma j}(t_0)$ (A.1) of each security in the portfolio.

At each time t_i during Δ (A.3), the investor may sum all $j=1,..J$ rescaled the values $c_j(t_i)$ of trades to obtain the value $Q(t_i)$, which models the trade with all J securities of his portfolio. The similar sum of all $j=1,..J$ rescaled volumes $u_j(t_i)$ of trades defines the volume $W(t_i)$ (A.12) of the trade with shares of all portfolio securities. The values $Q(t_i)$ and volumes $W(t_i)$ (A.12) of trades at time t_i define the prices $s(t_i)$ (A.12) per share of the portfolio. The total volume $W_\Sigma(t)$ (A.13) of trade during Δ (A.3) exactly equals the total number $W_\Sigma(t_0)$ (A.2) of shares in the portfolio at time t_0 in the past. The total value $Q_\Sigma(t)$ (A.14) of trade during Δ (A.3) defines the total current value of the portfolio. The total value $Q_\Sigma(t)$ (A.14) and volume $W_\Sigma(t)$ (A.13) of trade during Δ define the mean price $s(t)$ (B.1) per share of the portfolio during Δ (A.3) as volume weighted average price (VWAP) (Berkowitz et al., 1988).

The time series of values $Q(t_i)$, volumes $W(t_i)$, and prices $s(t_i)$ (A.12) model the trade with the portfolio as with a single security. The total trade volume $u_{\Sigma j}(t)$ (A.11) with shares of each security j during Δ exactly equals the number of shares $U_{\Sigma j}(t_0)$ (A.1) of security j in the portfolio at time t_0 in the past. The total trade volume $W_\Sigma(t)$ (A.13) equals the total number of shares $W_\Sigma(t_0)$ (A.2) in the portfolio. That describes the portfolio in the same way as the values $C_j(t_i)$, volumes $U_j(t_i)$, and prices $p_j(t_i)$ of trades describe the deals with a particular security j.

The mean price $s(t)$ (B.1) per share defines the mean return $R(t,t_0)$ (B.2) of the portfolio, like VWAP. The dependence of values $Q(t_i)$, volumes $W(t_i)$, and prices $s(t_i)$ (A.12) on rescaled values $c_j(t_i)$, volumes $u_j(t_i)$, and prices $p_j(t_i)$ (A.10-A.12) allows us to change the orders of sums and obtain the decomposition of portfolio mean return $R(t,t_0)$ by mean returns $R_j(t,t_0)$ of its securities (B.2), which, as it must be, coincides with one (1.1) derived by Markowitz (1952).

3. Market-Based Random Returns

The dependence of time series of values $Q(t_i)$, volumes $W(t_i)$, and prices $s(t_i)$ (A.12) on rescaled values $c_j(t_i)$, volumes $u_j(t_i)$, and prices $p_j(t_i)$ (A.10-A.12) defines the equation on portfolio random returns $R(t_i,t_0)$ (3.1; B.3) that accounts for random volumes $U_j(t_i)$ (A.4) of consecutive trades, random rescaled volumes $u_j(t_i)$ (A.10), and random volumes $W(t_i)$ (A.12):

$$R(t_i,t_0) = \sum_{j=1}^{J} R_j(t_i,t_0)\ X_j(t_0)\ \frac{x_j(t_i)}{x_j(t_0)} \qquad ; \qquad x_j(t_i) = \frac{u_j(t_i)}{W(t_i)} \tag{3.1}$$

We highlight that the differences of the equation on random returns $R(t_i,t_0)$ (3.1) from the equation (1.2; 1.3) proposed by Markowitz (1952) are determined by the impact of random coefficients $x_j(t_i)$ (3.1) that are equal to the ratios of rescaled random volumes $u_j(t_i)$ (A.10) of trades with security j to the random volumes $W(t_i)$ (A.12) that model the trades with all securities of the portfolio. The random coefficients $x_j(t_i)$ (3.1) account for random volumes $U_j(t_i)$ (A.4) of consecutive trades and that impact on the dependence of portfolio random returns $R(t_i,t_0)$ (3.1). Markowitz missed this dependence in his equation (1.2).

If the trade volumes $U_j(t_i)$ (A.4) with each portfolio security are constant during Δ, the equation on portfolio random returns $R(t_i,t_0)$ (3.1) coincides with Markowitz's equation (1.2; 1.3) (App. B., Olkhov, 2025c). That proves that Markowitz's equation on portfolio random returns $R(t_i,t_0)$ (1.2; 1.3) and Markowitz variance $\Theta_M(t,t_0)$ (1) describe only a limited case when all volumes of consecutive trades with all securities $j=1,..J$ in the portfolio are constant during Δ (A.3).

4. Market-Based Variance

We start with the derivation of market-based variance of portfolio random price $s(t_i)$ (A.12) that is determined by the values $Q(t_i)$ and volumes $W(t_i)$ (A.12) that model the trade with the portfolio as with a single security. The VWAP price $s(t)$ (B.1) of random prices $s(t_i)$ (A.12) follows from equations (A.14). To derive market-based variance $\Phi(t)$ (B.9) of random price

$s(t_i)$ (A.12), one should use the equation that determines the squares of prices $s^2(t_i)$ (B.4). The averaging of (B.4) is like VWAP (B.1) but is weighted by the squares of the trade volumes $W^2(t_i)$. The justification of market-based variance $\Phi(t)$ (B.9) of random price $s(t_i)$ (A.12) and its calculation are given by (B.4-B.17). Market-based variance $\Phi(t)$ of price $s(t_i)$ (A.12) per share takes the form (4.1; B.17):

$$\Phi(t) = \frac{\psi^2(Q) - 2\,\varphi(Q,W) + \chi^2(W)}{1 + \chi^2(W)}\, s^2(t) \tag{4.1}$$

The functions $\psi(Q)$, $\chi(W)$, and $\varphi(Q,W)$ in (4.1) denote the coefficients of variation of random values $Q(t_i)$, volumes $W(t_i)$, and their covariance (B.15). The market-based variance $\Theta(t,t_0)$ (B.18; 4.2) of returns is a direct consequence of the variance $\Phi(t)$ (4.1):

$$\Theta(t,t_0) = \frac{\Phi(t)}{s^2(t_0)} = \frac{\psi^2(Q) - 2\,\varphi(Q,W) + \chi^2(W)}{1 + \chi^2(W)}\, R^2(t,t_0) \tag{4.2}$$

The market-based variance of any security j has the same form as (4.2), but the mean return $R(t,t_0)$ and the coefficients of variation $\psi(Q)$, $\chi(W)$, and their covariance $\varphi(Q,W)$ in (B.15) should be determined by the time series of values $C_j(t_i)$ and volumes $U_j(t_i)$ of trades with a particular security j.

The portfolio market-based variance $\Theta(t,t_0)$ (4.2) reveals a dependence on random trade volumes $W(t_i)$ during Δ (A.3), which is described by the coefficient of variation $\chi(W)$ and covariance $\varphi(Q,W)$ (B.15). If one assumes that the trade volumes $U_j(t_i)$ with each security j of the portfolio, and hence, volumes $W(t_i)$ (A.12), are constant, then market-based variance $\Theta(t,t_0)$ (4.2) coincides with Markowitz variance $\Theta_M(t,t_0)$ (1), (B.19-B.25).

One may consider Markowitz variance $\Theta_M(t,t_0)$ (1) as a zero approximation of market-based variance $\Theta(t,t_0)$ (4.2) for the case $\chi(W)=0$. The expansion of market-based variance $\Theta(t,t_0)$ (4.2) into the Taylor series up to the 2^{nd} power by the coefficient of variation $\chi(W)$ with Markowitz variance $\Theta_M(t,t_0)$ (1) as a zero approximation $\Theta(t,t_0)|_{\chi=0} = \Theta_M(t,t_0)$ (1) is given in (Olkhov, 2025b). One should care that random trade volumes $W(t_i)$ may cause their coefficient of variation $\chi(W)$ to define the market-based variance $\Theta(t,t_0)$ (4.2) that is twice less or twice bigger than it is described by Markowitz variance $\Theta_M(t,t_0)$ (1) when $\chi(W)=0$ (Olkhov, 2025b).

The dependence of trade values $Q(t_i)$ and volumes $W(t_i)$ on the time series of rescaled values $c_j(t_i)$ and volumes $u_j(t_i)$ (A.12) determines the decompositions of the coefficients of variation $\psi(Q)$, $\chi(W)$, and covariance $\varphi(Q,W)$ (B.15) by the coefficients of variation $\psi_j(C)$, $\chi_j(U)$, and covariance $\varphi_{jk}(C,U)$ of their securities $j=1,..J$. That determines the decomposition of market-based variance $\Theta(t,t_0)$ (4.2) by the variances of portfolio securities (Olkhov, 2025a; 2025b).

5. Market-Based Sharpe Ratio

One may consider the equation (4.2) as a definition of the coefficient of variation $\mu(R)$ (5.1) of portfolio random returns $R(t_i,t_0)$ (3.1) that depends on the coefficients of variation of $\psi(Q)$, $\chi(W)$, and covariance $\varphi(Q,W)$ (B.15):

$$\mu^2(R) = \frac{\Theta(t,t_0)}{R^2(t,t_0)} = \frac{\psi^2(Q) - 2\,\varphi(Q,W) + \chi^2(W)}{1+\chi^2(W)} \tag{5.1}$$

However, since Sharpe (1966) introduced his ratio, practitioners much more use a Sharpe Ratio $Sp(R)$ (5.2) that is the reciprocal of the coefficient of variation $\mu(R)$ (5.1):

$$Sp^2(R) = \frac{R^2(t,t_0)}{\Theta(t,t_0)} = \frac{1+\chi^2(W)}{\psi^2(Q) - 2\,\varphi(Q,W) + \chi^2(W)} \tag{5.2}$$

We underline that we consider the Sharpe Ratio $Sp(R)$ (5.2) of gross returns $R(t,t_0)$ (A.7) instead of usual returns $r(t,t_0)= R(t,t_0)-1$. We highlight that the Sharpe Ratio $Sp(R)$ (5.2) accounts for random trade volumes and call it the market-based Sharpe Ratio. We underline that (5.2) determines the market-based Sharpe Ratio $Sp(R)$ of the portfolio and of any market security.

Actually, practitioners are more accustomed to using the Sharpe Ratio $Sp(r)$ (5.3) of the usual return $r(t,t_0)= R(t,t_0)-1$ (5.3) or the Sharpe Ratio $Sp(r_e)$ (5.4) of the excess return $r_e(t,t_0)$ (5.4), where r_f is a risk-free rate.

$$Sp(r) = \frac{r(t,t_0)}{\Theta^{1/2}(t,t_0)} = \frac{r(t,t_0)}{R(t,t_0)} \cdot Sp(R) \quad ; \quad r(t,t_0) = R(t,t_0) - 1 \tag{5.3}$$

$$Sp(r_e) = \frac{r(t,t_0) - r_f}{\Theta^{1/2}(t,t_0)} = \frac{r(t,t_0) - r_f}{R(t,t_0)} \cdot Sp(R) \quad ; \quad r_e(t,t_0) = r(t,t_0) - r_f \tag{5.4}$$

It is obvious that $Sp(R)$ completely describes both Sharpe Ratio $Sp(r)$ (5.3) and $Sp(r_e)$ (5.4). In App. C, we expand the Sharpe Ratio $Sp(R)$ (5.2) into a Taylor series (C.6) by the coefficient of variation $\chi(W)$ (B.15) of trade volumes up to its *2nd* power near $\chi=0$. When the trade volumes are assumed constant and $\chi=0$, the Sharpe Ratio equals $Sp_0(R)= \psi(Q)|_{\chi=0}$ (C6; C.7). In App. C, we present examples (C.8-C.1) that model the Sharpe Ratio $Sp(R)$ (5.2) for different values of $\chi(W)$ (B.15), $Sp_0(R)$, and coefficient of correlation a C(3) of trade values and volumes.

We recall that the dependence of the Sharpe Ratio $Sp(R)$ (5.2) on the coefficients of variation $\psi(Q)$, $\chi(W)$, and covariance $\varphi(Q,W)$ (B.15) reveals the impact of random trade volumes. To predict the market-based Sharpe Ratio $Sp(R)$ (5.2), one should forecast the time series of trade values and volumes, which is rather challenging.

6. Summary for Practitioners

1. Markowitz portfolio variance $\Theta_M(t,t_0)$ (1) describes a limited case of constant trade volumes. The implicit use of constant trade volumes approximation may be one of the reasons

for the low predictability of the mean and variance. Simply speaking, the use of the constant trade volumes while forecasting the mean and variance that in real markets depend on random trade volumes is very likely the use of constant steps while predicting random Brownian walks. That is not the best way to get the result. Market-based forecasts of the mean and variance of return of the portfolio at horizon T require predictions of the time series of random values and volumes of market trades with all securities of the portfolio at horizon T during the averaging interval. And that rises a completely different and very tough problem.

2. Practitioners should keep in mind that if they use only empirical time series of prices or returns for the assessments of their means and variances, then they implicitly use the limited case of constant trade volumes. The averaging procedures of prices and returns, which have the forms similar to (B.19-21; B.23), implicitly use the same limited assumption.

3. To estimate the means and variances that account for the random trade volumes, practitioners should use couples of empirical time series – values and volumes of trades, volumes and prices, or volumes and returns.

4. Practitioners may consider the coefficient of variation $\chi(W)$ (B.15) of trade volumes, as a measure of their randomness during Δ (A.3). If $\chi(W)$ is small enough, one may use the approximation of Markowitz's variance $\Theta_M(t,t_0)$ (1). The growth or decline of the market-based variance $\Theta(t,t_0)$ with increasing of $\chi(W)$ depends on coefficient of correlation a (C.3) of the trade values $Q(t_i)$ and volumes $W(t_i)$, and on the coefficient of variation $\psi(Q)$ (B.15) of trade values. (Olkhov, 2025b).

5. The Sharpe Ratio $Sp(R)$ (5.2) is the reciprocal to the coefficient of variation $\mu(R)$ (5.1) of returns. When all trade volumes are assumed constant the coefficient of variation $\chi(W)=0$ and the Sharpe Ratio $Sp_0(R)=\psi(Q)|_{\chi=0}$ (C.6; C.7). Practitioners may estimate the deviations of the market-based Sharpe Ratios $Sp(R)$ (5.3; 5.4; C.6-C.10) from $Sp_0(R)$ (App. C.).

6. Practitioners should check the models and methods they use while taking their investment decisions to explicitly approve or change their implicit approximations. In particular, if investment decisions depend on "empirical estimates" of price-volume correlations, they might be reconsidered. "Empirical assessments" of price-volume correlations are another example of the implicit and improper use of constant trade volumes approximation. That is a persuasive lesson that at least sometimes the investment and market intuition of practitioners, even when "supported" by empirical time series, must follow the proper theoretical considerations. Actually, the irregular or random time series of trade values and

volumes defines the mean price $p_j(t)$, which has the form of VWAP (A.6) and which directly determines zero price-volume correlations (D.4; App. D).

7. The "empirical assessments" of non-zero price-volume correlations result from the improper use of the mean prices $\pi_j(t)$ (B.19), which simply equal VWAP $p_j(t)$ (A.6) in the case of the constant trade volumes (D.5-D.7). The case of *constant trade volumes* is inconsistent with the intent to "empirically calculate" the correlation between *random trade volumes* and prices. Actually, the "empirical calculations" of price-volume correlations calculate the difference (D.6) between VWAP $p_j(t)$ (A.6) and $\pi_j(t)$ (B.19). Practitioners, who search for the statistical dependence between random trade volumes and prices, should empirically assess correlations between random prices and ***squares*** of trade volumes. In App. D, we give the correct market-based procedure for empirical calculations of these correlations.

8. The assessments of means-variances of the entire market trades at current time t may incorrectly estimate the current performance of market portfolios composed at time t_0 in the past. These deviations have their origin in the distinctions between the initial proportions of the amounts invested in market portfolios at time t_0 in the past and the random proportions of market trades at current time t with market securities of the entire market (Olkhov, 2025d).

9. All methods for optimal portfolio selection use the forecasts of means and variances of the securities at a certain time horizon T. The reliability and accuracy of these forecasts determine the reliability of portfolio selection. Practitioners should care that the uncertainty of time series of random trades and economic complexity limit the accuracy of any forecasts of prices and returns at best by the accuracy of Gaussian distributions (Olkhov, 2023a; 2023b; 2024). The dependence of market-based means and variances of securities and portfolios on the time series of values and volumes of successive trades results in the conclusion that their reliable predictions at horizon T depend on the forecasts of the corresponding time series of values and volumes of trades at the same horizon T during the averaging interval Δ (A.3). The predictions of trade time series are interlinked with the forecasts of the macroeconomic variables and environment. The forecasting of the mutual dependence between market trade and the macroeconomic environment is challenging even for BlackRock's Aladdin. Actually, we see the need and may propose the methods for the development of *"Divine Augur,"* which could greatly enhance Aladdin's prediction capacity.

7. Conclusion

This paper studies the portfolio variance determined by random time series of values and volumes of consecutive trades made daily at the exchange and doesn't account for the impact of annual dividend payoffs. The random time series of values and volumes of market trades with securities determine their means-variances. Markowitz's variance $\Theta_M(t,t_0)$ (1) describes a limited case when the volumes of consecutive trades are assumed constant. To overcome Markowitz's limitation, we show how to convert the time series of trades made at the exchange with portfolio securities and derive the time series that model the trades with the portfolio as with a single security.

That establishes an equal description of the variance for any securities and portfolios. The coefficient of variation $\chi(W)$ (B.15) of trade volumes may serve as a measure of their randomness. If $\chi(W)$ is small enough, one may use Markowitz variance $\Theta_M(t,t_0)$. We derive the Taylor series of the market-based variance $\Theta(t,t_0)$ (Olkhov, 2025b) and the Sharpe Ratio $Sp(R)$ (5.2; C.6-C.10) up to the 2[nd] power of the coefficient of variation $\chi(W)$ of random trade volumes that use Markowitz variance $\Theta_M(t,t_0)$ (1) and the Sharpe Ratio $Sp_0(R)$ (C6; C.7) as zero approximations when $\chi(W)=0$.

Practitioners should check the methods that support their investment decisions and approve or change the implicit use of constant trade volumes assumptions. Practitioners, who search for the statistical dependence between random trade volumes and prices, should empirically assess correlations between random prices and ***squares*** of trade volumes.

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We thank Prof. Sanjiv Das for his proposal to consider the Sharpe Ratio as a useful application of the market-based variance and for his advice to simplify the text for practitioners.

Appendix A. Trade with portfolio as with a single security

Let us consider the investor who at time t_0 in the past collected his portfolio with $U_{\Sigma j}(t_0)$ shares of each security j, $j=1,..J$. At time t_0 the volume of $U_{\Sigma j}(t_0)$ shares of security j had the value $C_{\Sigma j}(t_0)$ and the price $p_j(t_0)$, so they obey equation (A.1):

$$C_{\Sigma j}(t_0) = p_j(t_0) \cdot U_{\Sigma j}(t_0) \tag{A.1}$$

The total value $Q_\Sigma(t_0)$ and the number $W_\Sigma(t_0)$ of shares in the portfolio equal:

$$Q_\Sigma(t_0) = \sum_{j=1}^{J} C_{\Sigma j}(t_0) \quad ; \quad W_\Sigma(t_0) = \sum_{j=1}^{J} U_{\Sigma j}(t_0) \quad ; \quad Q_\Sigma(t_0) = s(t_0) W_\Sigma(t_0) \tag{A.2}$$

The equation (A.2) introduces the price $s(t_0)$ per share of the portfolio at time t_0. At current time t the investor doesn't trade the shares of his portfolio but during the averaging interval Δ (A.3) observes the time series of the values $C_j(t_i)$ and volumes $U_j(t_i)$ of consecutive trades made in the market with each of his securities $j=1,..J$ at time t_i.

$$\Delta = \left[t - \frac{\Delta}{2};\ t + \frac{\Delta}{2}\right] \quad ; \quad t_i \in \Delta \quad ; \quad i = 1, \dots N \quad ; \quad N \cdot \varepsilon = \Delta \tag{A.3}$$

We assume that the time span ε between successive trades at time t_i and $t_{i+1}=t_i+\varepsilon$ is constant and the same for all securities in the portfolio. The intervals between consecutive trades at a modern exchange is less than 1 second. It is not easy to collect and process time series of the values $C_j(t_i)$ and volumes $U_j(t_i)$ with all securities of the portfolio with such a high frequency. Actually, the exchange or the investors may sum the values $C_j(t_i)$ and volumes $U_j(t_i)$ of consecutive trades made at the exchange during any reasonable period that may be equal to 1 min, 1 hour, 1 day, or whatever. Further in this paper, as time series of the values $C_j(t_i)$ and volumes $U_j(t_i)$ of consecutive trades, we consider the sums of the values $C_j(t_i)$ and volumes $U_j(t_i)$ of trades during a particular period ε that can be equal to 1 min, 1 hour, 1 day, etc. The period ε determines aggregation of the initial trade time series. The choice of the duration of the period ε is determined by the problem under consideration. For example, if portfolio managers are looking for mean and variance during the averaging interval Δ that equals a trading day (~ 6-8 hours), they should choose the period $\varepsilon<<\Delta$, for example $\varepsilon \sim 5\ min$, to obtain 70-100 terms of the time series of consecutive trades during the interval Δ. Ultimately, as consecutive trades, we consider the sums of the values and volumes of trades during the selected period ε. We average the consecutive trades during the interval Δ, and $\varepsilon<<\Delta$.

The values $C_j(t_i)$ and volumes $U_j(t_i)$ of trades define the prices $p_j(t_i)$ (A.4) of each security j:

$$C_j(t_i) = p_j(t_i) \cdot U_j(t_i) \tag{A.4}$$

The mean value $C_j(t)$, mean volume $U_j(t)$, the total value $C_{\Sigma j}(t)$, and the total volume $U_{\Sigma j}(t)$ of trades with security j at current time t during the interval Δ (A.3) equal:

$$C_j(t) = \frac{1}{N}\sum_{i=1}^{N} C_j(t_i) \quad ; \ C_{\Sigma \mathrm{j}}(t) = N \cdot C_j(t)\,; \ \ U_{\mathrm{j}}(t) = \frac{1}{N}\sum_{i=1}^{N} U_j(t_i)\ ; \ U_{\Sigma \mathrm{j}}(t) = N \cdot U_j(t) \tag{A.5}$$

The mean price $p_j(t)$ (A.6) of the trades with shares of security j at time t during Δ takes the well-known form of volume weighted average price (VWAP) (Berkowitz et al., 1988):

$$p_j(t) = \frac{C_{\Sigma \mathrm{j}}(t)}{U_{\Sigma \mathrm{j}}(t)} = \frac{C_{\mathrm{j}}(t)}{U_{\mathrm{j}}(t)} = \frac{1}{U_{\Sigma \mathrm{j}}(t)}\sum_{i=1}^{N} p_j(t_i)\, U_j(t_i) \tag{A.6}$$

The mean return $R_j(t,t_0)$ (A.7) of security j with respect to its price $p_j(t_0)$ (A.1) at time t_0:

$$R_j(t,t_0) = \frac{p_j(t)}{p_j(t_0)} = \frac{1}{U_{\Sigma \mathrm{j}}(t)}\sum_{i=1}^{N} R_j(t,t_0)\, U_j(t_i) \quad ; \quad R_j(t,t_0) = \frac{p_j(t_i)}{p_j(t_0)} \tag{A.7}$$

We use "gross" return $R_j(t,t_0)$ (A.7) instead of the usual return $r_j(t,t_0)= R_j(t,t_0)-1$, but the variance of both returns is the same. $R_j(t_i,t_0)$ (A.7) at time t_i defines random returns of trades (A.4) with security j. Markowitz (1952) determined the mean return $R(t,t_0)$ of the portfolio:

$$R(t,t_0) = \sum_{j=1}^{J} R_j(t,t_0) \cdot X_j(t_0) \quad ; \quad X_j(t_0) = \frac{C_{\Sigma j}(t_0)}{Q_{\Sigma}(t_0)} \tag{A.8}$$

$X_j(t_0)$ in (A.8) is a relative amount invested into security j at time t_0 in the past. The equation (A.8) describes the dependence of portfolio mean return $R(t,t_0)$ on the mean returns $R_j(t,t_0)$ (A.7) of its securities. Markowitz (1952, p. 81) derived his equation (A.9) on portfolio random returns $R(t_i,t_0)$ without justification but simply took it in the same form as the equation (A.8):

$$R(t_i,t_0) = \sum_{j=1}^{J} R_j(t_i,t_0)\, X(t_0) \tag{A.9}$$

The average of the equation (A.9) determines the dependence of portfolio mean return $R(t,t_0)$ (A.8) on the mean returns $R_j(t,t_0)$ of its securities. On the other hand, the equation (A.9) immediately defines the expression of Markowitz's variance $\Theta_M(t,t_0)$ (1) (Markowitz, 1952, p. 81). For the last 75 years, the equation on random returns $R(t_i,t_0)$ (A.9) served as the solid ground for the assessments of the portfolio variances and risks.

Meanwhile, the equation (A.8) on mean returns can't exactly determine the dependence between random returns $R(t_i,t_0)$ (A.9) of the portfolio and its securities $R_j(t_i,t_0)$ because many probabilities may have the same mean values (Shiryaev, 1999). We show below that Markowitz's equation on random returns (A.9) is correct only for a limited case when the volumes of all trades with portfolio securities are assumed to be constant during Δ (A.3).

To show that and to overcome the limitation of Markowitz's equation (A.9), the investor should convert the observed time series of trades with his securities in the market to obtain the time series that model the trades with his portfolio as with a single security. To do that, let us multiply each value $C_j(t_i)$ and volume $U_j(t_i)$ of trades at time t_i with security j by the ratio of the number of shares $U_{\Sigma j}(t_0)$ of security j in his portfolio at time t_0 to the total volume $U_{\Sigma j}(t)$ (A.4) of trades with security j during Δ. We obtain rescaled values $c_j(t_i)$ and volumes $u_j(t_i)$:

$$c_j(t_i) = \frac{U_{\Sigma j}(t_0)}{U_{\Sigma j}(t)} \cdot C_j(t_i) \quad ; \quad u_j(t_i) = \frac{U_{\Sigma j}(t_0)}{U_{\Sigma j}(t)} \cdot U_j(t_i) \quad ; \quad c_j(t_i) = p_j(t_i) \cdot u_j(t_i) \tag{A.10}$$

The rescaled values $c_j(t_i)$ and volume $u_j(t_i)$ of trades obey equation (A.10) with the same price $p_j(t_i)$ (A.4). For each security $j=1,..J$ in the portfolio, the time series of the rescaled values $c_j(t_i)$ and volumes $u_j(t_i)$ (A.10) model the trade with the total volume $u_{\Sigma j}(t)$ (A.11) during Δ, which exactly equals the number $U_{\Sigma j}(t_0)$ of shares of security j in the portfolio at time t_0 in the past:

$$u_{\Sigma j}(t) = \sum_{i=1}^{N} u_j(t_i) = \frac{U_{\Sigma j}(t_0)}{U_{\Sigma j}(t)} \sum_{i=1}^{N} U_j(t_i) = \frac{U_{\Sigma j}(t_0)}{U_{\Sigma j}(t)} U_{\Sigma j}(t) = U_{\Sigma j}(t_0) \tag{A.11}$$

To define the time series of the values $Q(t_i)$ and volumes $W(t_i)$ (A.12) of trades, which model the trade with the portfolio as with a single security, let us at time t_i sum the rescaled values $c_j(t_i)$ and volumes $u_j(t_i)$ (A.10) of all securities $j=1,..J$ in the portfolio:

$$Q(t_i) = \sum_{j=1}^{J} c_j(t_i) \quad ; \quad W(t_i) = \sum_{j=1}^{J} u_j(t_i) \quad ; \quad Q(t_i) = s(t_i) \cdot W(t_i) \tag{A.12}$$

The equation (A.12) models the trade at price $s(t_i)$ per share of the portfolio at time t_i during Δ (A.3). At current time t the total trade volume $W_\Sigma(t)$ (A.13) equals the number $W_\Sigma(t_0)$ (A.2) of shares in the portfolio at time t_0 in the past.

$$W_\Sigma(t) = \sum_{i=1}^{N} W(t_i) = \sum_{j=1}^{J} \sum_{i=1}^{N} u_j(t_i) = \sum_{j=1}^{J} U_j(t_0) = W_\Sigma(t_0) \qquad \text{(A.13)}$$

The total value $Q_\Sigma(t)$ (A.14) of the portfolio at current time t during Δ determines the mean price $s(t)$ (A.14) per share of the portfolio:

$$Q_\Sigma(t) = \sum_{i=1}^{N} Q(t_i) \quad ; \quad Q_\Sigma(t) = s(t) W_\Sigma(t) \quad ; \quad s(t) = \frac{1}{W_\Sigma(t_0)} \sum_{i=1}^{N} s(t_i)\, W(t_i) \qquad \text{(A.14)}$$

The time series of the values $Q(t_i)$, volumes $W(t_i)$, and prices $s(t_i)$ (A.12) model the trades with each security $j=1,..J$ of the portfolio in the amounts that are equal to the numbers $U_{\Sigma j}(t_0)$ of shares of its securities. The total trade volume $W_\Sigma(t)$ (A.13) of trades with the portfolio exactly equals the total number of its shares $W_\Sigma(t_0)$ (A.2) at time t_0 in the past. Thus, one may conclude that the time series of the values $Q(t_i)$, volumes $W(t_i)$, and prices $s(t_i)$ (A.12) model the trade with the portfolio as with a single security. We recall that at current time t the investor doesn't trade his shares, and the time series $Q(t_i)$, $W(t_i)$, and $s(t_i)$ (A.12) only simulate and model the deals with the portfolio and give the assessments of the portfolio mean return and the variance.

Appendix B. Random returns, mean returns, and market-based variance

The dependence of the time series of the values $Q(t_i)$, volumes $W(t_i)$, and prices $s(t_i)$ (A.12) on the time series of rescaled values $c_j(t_i)$, volumes $u_j(t_i)$, and prices $p_j(t_i)$ (A.10) and (A.4; A.6) determines the decomposition of the mean price $s(t)$ (A.14) per share of the portfolio by the mean prices $p_j(t)$ (A.6) of portfolio securities $j=1,..J$:

$$s(t) = \frac{Q_\Sigma(t)}{W_\Sigma(t_0)} = \frac{\sum_{i=1}^{N} s(t_i) W(t_i)}{W_\Sigma(t_0)} = \frac{1}{W_\Sigma(t_0)} \sum_{j=1}^{J} C_{\Sigma j}(t) \frac{U_{\Sigma j}(t_0)}{U_{\Sigma j}(t)} = \sum_{j=1}^{J} p_j(t) x_j(t_0) \;\; ; \;\; x_j(t_0) = \frac{U_{\Sigma j}(t_0)}{W_\Sigma(t_0)} \qquad \text{(B.1)}$$

Coefficients $x_j(t_0)$ in (B.1) define the relative number of shares of security j in the portfolio at time t_0. The decomposition of the mean price $s(t)$ (B.1) of the portfolio determines the decomposition of the mean return $R(t,t_0)$ (B.2) of the portfolio with respect to its price $s(t_0)$ (A.2) by the mean returns $R_j(t,t_0)$ (B.2; A.7) of its securities:

$$R(t,t_0) = \frac{s(t)}{s(t_0)} = \sum_{j=1}^{J} \frac{p_j(t) x_j(t_0)}{s(t_0)} \sum_{j=1}^{J} \frac{p_j(t)}{p_j(t_0)} \cdot \frac{p_j(t_0) \cdot U_{\Sigma j}(t_0)}{s(t_0) \cdot W_\Sigma(t_0)} = \sum_{j=1}^{J} R_j(t,t_0) \cdot X_j(t_0) \qquad \text{(B.2)}$$

The decomposition of the mean return $R(t,t_0)$ (B.2) completely coincides with Markowitz's expression (3.6; A.8). However, the dependence of $Q(t_i)$, $W(t_i)$, and $s(t_i)$ (A.12) on $c_j(t_i)$, $u_j(t_i)$, and $p_j(t_i)$ (A.10) reveals that the equation on random returns (B.3) differs from Markowitz's equation (A.9). From (A.12), obtain the expression for portfolio random returns $R(t_i,t_0)$:

$$R(t_i,t_0)=\frac{s(t_i)}{s(t_0)}=\frac{Q(t_i)}{s(t_0)W(t_i)}=\frac{\sum_{j=1}^{J}c_j(t_i)}{s(t_0)W(t_i)}=\frac{\sum_{j=1}^{J}p_j(t_i)\,u_j(t_i)}{s(t_0)W(t_i)}=\sum_{j=1}^{J}\frac{p_j(t_i)}{p_j(t_0)}\frac{p_j(t_0)U_j(t_0)}{s(t_0)W_\Sigma(t_0)}\frac{u_j(t_i)}{W(t_i)}\frac{W_\Sigma(t_0)}{U_j(t_0)}$$

Simple transformations give the equation on the portfolio random return $R(t_i,t_0)$ (B.3) that accounts for random relative volumes $x_j(t_i)$ (B.3) of the consecutive trades, which was missed by Markowitz's equation (A.9):

$$R(t_i,t_0)=\sum_{j=1}^{J}R_j(t_i,t_0)\cdot X_j(t_0)\cdot\frac{x_j(t_i)}{x_j(t_0)}\quad;\quad x_j(t_i)=\frac{u_j(t_i)}{W(t_i)}\qquad\text{(B.3)}$$

Let us show that the equation (A.8) on portfolio mean return $R(t,t_0)$ follows from (B.3). From the definition of mean return (A.7) and (B.3), obtain:

$$R(t,t_0)=\frac{1}{W_\Sigma(t_0)}\sum_{i=1}^{N}R(t_i,t_0)\,W(t_i)=\frac{1}{W_\Sigma(t_0)}\sum_{j=1}^{J}X_j(t_0)\cdot\sum_{i=1}^{N}R_j(t_i,t_0)\cdot\frac{u_j(t_i)}{W(t_i)}\cdot\frac{W_\Sigma(t_0)}{U_{\Sigma j}(t_0)}\cdot W(t_i)$$

The use of the rescaled volumes $u_j(t_i)$ (A.10) gives:

$$R(t,t_0)=\sum_{j=1}^{J}X_j(t_0)\cdot\frac{1}{U_{\Sigma j}(t_0)}\sum_{i=1}^{N}R_j(t_i,t_0)\frac{U_{\Sigma j}(t_0)}{U_{\Sigma j}(t)}\cdot U_j(t_i)=\sum_{j=1}^{J}R_j(t,t_0)\cdot X_j(t_0)$$

The above equation coincides with Markowitz' equation (A.8) and proves that (B.3) correctly describes portfolio mean returns.

One can easily derive (Olkhov, 2025c) that if for each security $j=1,..J$ all trade volumes $U_j(t_i)$ are assumed to be constant, then, from (A.5; A.10; A.11; A.13) and (B.1; B.3), obtain :

$$U_j(t_i)=const\ \rightarrow U_j(t_i)=\frac{U_{\Sigma j}(t)}{N}\ ;\ u_j(t_i)=\frac{U_{\Sigma j}(t_0)}{N}\ ;\ W(t_i)=\frac{W_\Sigma(t_0)}{N}\rightarrow x_j(t_i)=x_j(t_0)$$

Hence, $x_j(t_i)/x_j(t_0)$ =1, and the equation on random returns $R(t_i,t_0)$ (B.3) coincides with Markowitz's equation (A.9). That proves that Markowitz's equation on random returns (A.9) and Markowitz variance $\Theta_M(t,t_0)$ (1), which follows from (A.9), describe only the limited case when the trade volumes of all portfolio securities are assumed to be constant.

The time series of the values $Q(t_i)$, volumes $W(t_i)$, and prices $s(t_i)$ (A.12) that model the trade with the portfolio as with a single security allows us to derive market-based portfolio variance that accounts for random volumes $W(t_i)$ of the consecutive trades with the portfolio during Δ (A.3). The market-based variance of the portfolio that consists of $j=1,..J$ securities has the same form as the market-based variance of any single security.

We start with the introduction of equation (B.4) that defines the squares of price $s^2(t_i)$:

$$Q^2(t_i)=s^2(t_i)\cdot W^2(t_i)\qquad\text{(B.4)}$$

The equation (B.4) is a square of the equation (A.10). The N terms of the time series of the values $Q(t_i)$ and volumes $W(t_i)$ approximate their means (B.5), mean squares (B.6), and their joint mathematical expectation $E[Q(t_i)W(t_i)]$ (B.7):

$$E[Q(t_i)]=Q(t)=\frac{1}{N}\sum_{i=1}^{N}Q(t_i)\quad;\quad E[W(t_i)]=W(t)=\frac{1}{N}\sum_{i=1}^{N}W(t_i)\qquad\text{(B.5)}$$

$$Q(t;2) = \frac{1}{N}\sum_{i=1}^{N} Q^2(t_i) \ \ ; \ \ W(t;2) = \frac{1}{N}\sum_{i=1}^{N} W^2(t_i) \tag{B.6}$$

$$E[Q(t_i)W(t_i)] = \frac{1}{N}\sum_{i=1}^{N} Q(t_i)\, W(t_i) \tag{B.7}$$

Similar to (A.13; A.14), define the total sum of squares of values $Q_\Sigma(t;2)$, mean squares of values $Q(t;2)$, total sum of squares of volumes $W_\Sigma(t;2)$, and mean squares of volumes $W_\Sigma(t;2)$:

$$Q_\Sigma(t;2) = \sum_{i=1}^{N} Q^2(t_i) = N \cdot Q(t;2) \quad ; \quad W_\Sigma(t;2) = \sum_{i=1}^{N} W^2(t_i) = N \cdot W(t;2) \tag{B.8}$$

We use (B4-B.8) and define market-based variance $\Phi(t)$ (B.9) of price $s(t_i)$ (A.12) that accounts for random squares of volumes $W^2(t_i)$ of trades during Δ (A.3) (Olkhov, 2025a-2025d):

$$\Phi(t) = \frac{1}{W_\Sigma(t;2)}\sum_{i=1}^{N}\big(s(t_i) - s(t)\big)^2\, W^2(t_i) \tag{B.9}$$

The averaging in (B.9) is like VWAP (A.6) but is weighted by the squares $W^2(t_i)$ of trade volumes. Let us present (B.9) as:

$$\Phi(t) = \Phi(t;2) - 2\Phi(t;1)s(t) + s^2(t) \tag{B.10}$$

The use of (B.4-B.8) presents the function $\Phi(t;2)$ as (B.11):

$$\Phi(t;2) = \frac{1}{W_\Sigma(t;2)}\sum_{i=1}^{N} s^2(t_i)\, W^2(t_i) = \frac{\sum_{i=1}^{N} Q^2(t_i)}{W_\Sigma(t;2)} = \frac{Q_\Sigma(t;2)}{W_\Sigma(t;2)} = \frac{Q(t;2)}{W(t;2)} \tag{B.11}$$

The use of (B.4; B.6-B.8) presents the function $\Phi(t;1)$ (B.10) as follows:

$$\Phi(t;1) = \frac{1}{W_\Sigma(t;2)}\sum_{i=1}^{N} s(t_i)\, W^2(t_i) = \frac{1}{W(t;2)}\,\frac{1}{N}\sum_{i=1}^{N} Q(t_i)\, W(t_i) = \frac{E[Q(t_i)\, W(t_i)]}{W(t;2)} \tag{B.12}$$

The use of properties of the mean squares $Q(t;2)$, $W(t;2)$ (B.6), and of the joint mathematical expectation $E[Q(t_i)W(t_i)]$ (B.7) (Shiryaev, 1999), present them as the coefficients of variations:

$$Q(t;2) = Q^2(t)(1 + \psi^2(Q)) \quad ; \quad W(t;2) = W^2(t)[1 + \chi^2(W)] \tag{B.13}$$

$$E[Q(t_i)\, W(t_i)] = Q(t)W(t)(1 + \varphi(Q,W)\} \tag{B.14}$$

In (B.13; B.14) we introduce the coefficient of variation $\psi(Q)$ (B.15) of the values $Q(t_i)$, the coefficient of variation $\chi(W)$ (B.15) of the volumes $W(t_i)$, and their covariance $\varphi(Q,W)$ (B.15) normalized to their mean values $Q(t)$ and volumes $W(t)$ (B.5):

$$\psi^2(Q) = \frac{cov\{Q(t),Q(t)\}}{Q^2(t;1)} \quad ; \quad \chi^2(W) = \frac{cov\{W(t),W(t)\}}{W^2(t;1)} \quad ; \quad \varphi(Q,W) = \frac{cov\{Q(t),W(t)\}}{Q(t;1)W(t;1)} \tag{B.15}$$

The covariances of $Q(t_i)$ and $W(t_i)$ in (B.15) have the usual definitions (B.16).

$$cov\{Q(t),W(t)\} = \frac{1}{N}\sum_{i=1}^{N}[Q(t_i) - Q(t)] \cdot [W(t_i) - W(t)] \tag{B.16}$$

All averaging (B.5-B.7; B.12; B.16) gives the approximations of mathematical expectations by the finite number N of terms of the time series during Δ (A.3). After the transformations and the use of (B.13-B.16), we convert the market-based variance $\Phi(t)$ (B.9; B.10) of price $s(t)$ that accounts for random volumes $W(t_i)$ of market trades into (B.17) (Olkhov, 2025a-2025d):

$$\Phi(t) = \frac{\psi^2(Q) - 2\,\varphi(Q,W) + \chi^2(W)}{1 + \chi^2(W)}\, s^2(t) \tag{B.17}$$

The market-based variance $\Phi(t)$ (B.17) determines market-based variance $\Theta(t,t_0)$ of returns:

$$\Theta(t,t_0) = \frac{\Phi(t)}{s^2(t_0)} = \frac{\psi^2(Q)-2\,\varphi(Q,W)+\chi^2(W)}{1+\chi^2(W)}\ R^2(t,t_0) \tag{B.18}$$

The decomposition of the market-based variance $\Theta(t,t_0)$ (B.18) by its securities follows from the decomposition of the mean return $R(t,t_0)$ (A.8) by mean returns $R_j(t,t_0)$ of its securities and from the dependence of the time series of the values $Q(t_i)$, volumes $W(t_i)$, and prices $s(t_i)$ (A.12) on time series of rescaled values $c_j(t_i)$, volumes $u_j(t_i)$, and prices $p_j(t_i)$ of portfolio securities. That determines the corresponding dependence of the coefficients of variation $\psi(Q)$, $\chi(W)$, and the covariance $\varphi(Q,W)$ (B.15) of the portfolio on the coefficients of variation and covariances of securities that compose the portfolio (Olkhov, 2025a; 2025b).

If all volumes $U_j(t_i)$-*const*, then mean prices (A.6) takes the form:

$$p_j(t)|_{U_j(t_i)-const} = \frac{1}{U_{\Sigma j}(t)}\sum_{i=1}^{N} p_j(t_i)\,U_j(t_i)|_{U_j(t_i)-const} = \frac{1}{N}\sum_{i=1}^{N} p_j(t_i) = \pi_j(t) \tag{B.19}$$

If all volumes $U_j(t_i)$-*const*, the mean return $R_j(t,t_0)$ (A.7) of security j takes the form:

$$R_j(t,t_0) = \frac{1}{U_{\Sigma j}(t)}\sum_{i=1}^{N} R_j(t,t_0)\,U_j(t_i)|_{U_j(t_i)-const} = \frac{1}{N}\sum_{i=1}^{N} R_j(t,t_0) \tag{B.20}$$

If all volumes $W(t_i)$-*const*, the mean price $s(t)$ (B.1) per share of the portfolio takes the form:

$$s(t) = \frac{\sum_{i=1}^{N} s(t_i)W(t_i)}{W_\Sigma(t_0)} = \frac{1}{N}\sum_{i=1}^{N} s(t_i) \tag{B.21}$$

If $W(t_i)$-*const*, then the coefficient of variation $\chi(W)=0$ and covariance $\varphi(Q,W)=0$. Then:

$$\frac{\psi^2(Q)-2\,\varphi(Q,W)+\chi^2(W)}{1+\chi^2(W)}|_{\chi=0} = {\psi_0}^2(Q) = \frac{cov\{Q(t),Q(t)\}}{Q^2(t;1)}|_{\chi=0} = \frac{cov\{s(t),s(t)\}}{s^2(t)} \tag{B.22}$$

$$cov\{s(t),s(t)\}|_{\chi=0} = \frac{1}{N}\sum_{i=1}^{N}\big(s(t_i)-s(t)\big)^2 \tag{B.23}$$

$$\Theta(t,t_0)|_{\chi=0} = \Theta_c(t,t_0) = \frac{1}{N}\frac{\sum_{i=1}^{N}\big(s(t_i)-s(t)\big)^2}{s^2(t)}\frac{s^2(t)}{s^2(t_0)} = \frac{1}{N}\sum_{i=1}^{N}\frac{\big(s(t_i)-s(t)\big)^2}{s^2(t_0)} = \frac{1}{N}\sum_{i=1}^{N}\big(R(t_i,t_0)-R(t,t_0)\big)^2$$

The use of equations (1.1; 1.3) and the change of the order of sums give:

$$\Theta_c(t,t_0) = \sum_{j,k=1}^{J} X_j(t_0)X_k(t_0)\cdot\frac{1}{N}\sum_{i=1}^{N}\Big(R_j(t_i,t_0)-R_j(t,t_0)\Big)\big(R_k(t_i,t_0)-R_k(t,t_0)\big)$$

Finally, obtain that if $\chi(W)=0$, then market-based variance $\Theta_c(t,t_0)$ (B.24) coincides with Markowitz variance $\Theta_M(t,t_0)$ (1), which ignores the impact of random trade volumes:

$$\Theta_c(t,t_0) = {\psi_0}^2(Q)R^2(t,t_0) = \sum_{j,k=1}^{J}\theta_{jk}(t,t_0)X_j(t_0)\,X_k(t_0) = \Theta_M(t,t_0) \tag{B.24}$$

$$\theta_{jk}(t,t_0) = \frac{1}{N}\sum_{i=1}^{N}\Big(R_j(t_i,t_0)-R_j(t,t_0)\Big)\big(R_k(t_i,t_0)-R_k(t,t_0)\big) \tag{B.25}$$

Appendix C. Market-based Sharpe Ratio

Consider the expansion of the market-based Sharpe Ratio $Sp(R)$ (C.1; 5.2) in the Taylor series. For brevity we omit the dependence on time t of ψ, χ, and φ (B.15)

$$Sp^2(R) = \frac{R^2(t,t_0)}{\Theta(t,t_0)} = \mu^{-2}(R) = \left[\frac{1+\chi^2}{\psi^2-2\,\varphi+\chi^2}\right]^{1/2} \tag{C.1}$$

The use of the Cauchy-Schwarz-Bunyakovskii inequality (Shiryaev, 1999, p 123; Olkhov, 2025a; 2025b) for the covariance φ (B.15), gives:

$$|\varphi| = \left|\frac{cov\{Q(t),W(t)\}}{Q(t;1)W(t;1)}\right| \le \psi\,\chi \tag{C.2}$$

The inequality (C.2) presents the covariance φ as (C.3):

$$\varphi = \frac{cov\{Q(t),W(t)\}}{Q(t;1)W(t;1)} = a\cdot\psi\cdot\chi \quad ; \quad -1 \le a \le 1 \tag{C.3}$$

The coefficient a (C.3) is the coefficient of correlation of trade values $Q(t_i)$ and volumes $W(t_i)$. The change of variables $x=\chi/\psi$ converts the Sharpe Ratio $Sp(R)$ (C.1) into (C.4):

$$\chi = \psi\,x \quad \rightarrow \quad Sp(R) = \left[\frac{1+\chi^2}{\psi^2-2\,\varphi+\chi^2}\right]^{1/2} = \frac{1}{\psi}\left[\frac{1+\psi^2x^2}{1-2\,a\cdot x+x^2}\right]^{1/2} = \frac{1}{\psi}\cdot f(x) \tag{C.4}$$

The Taylor series of the function $f(x)$ up to the 2nd power of x^2 takes the form:

$$f(x) = \left[\frac{1+\psi^2x^2}{1-2\,a\cdot x+x^2}\right]^{1/2} \sim 1 + a\cdot x + \frac{1}{2}[3a^2 - (1-\psi_0^2)]\cdot x^2 \quad ; \quad \psi_0 = \psi|_{\chi=0} \tag{C.5}$$

From (B.22) and (C.4; C.5), obtain the Taylor series of the Sharpe Ratio $Sp(R)$ (C.6):

$$Sp(R) = Sp_0(R)\left[1 + aSp_0(R)\cdot\chi + \frac{1}{2}[3a^2Sp_0^2(R) + \left(1-Sp_0^2(R)\right)]\cdot\chi^2\right] \; ; \; Sp_0(R) = \psi_0^{-1} \tag{C.6}$$

We denote $Sp_0(R)$ (C.6; C.7) as the Sharpe Ratio when trade volumes $W(t_i)=W$ are constant and $\chi(W)=0$. Then the mean price $s(t)$ (B.21), the square of the coefficient of variation $\psi_0^2(Q)$ (B.22), and the Sharpe Ratio $Sp_0(R)$ take the form:

$$Sp^2(R)|_{\chi=0} = Sp_0^2(R) = \psi_0^{-2} = \frac{s^2(t)}{\frac{1}{N}\sum_{i=1}^{N}\left(s(t_i)-s(t)\right)^2} \tag{C.7}$$

The Taylor series shows the change of the Sharpe Ratio $Sp(R)$ (C.6) with the increasing of the coefficient of variation $\chi(W)$ (B.15) depending on the coefficient of correlation a (C.3) and the value of the Sharpe Ratio $Sp_0(R)$ (C.7).

1. If the coefficient of correlation $a>0$ (C.3) and $Sp_0(R)<1$, the Sharpe Ratio $Sp(R)$ (C.6) grows up with increasing $\chi(W)$ from zero. For example, if a, $Sp_0(R)$, and χ are equal (C.8):

$$a\sim\frac{1}{2} \quad ; \quad Sp_0(R) = \psi_0^{-1}\sim 2 \; ; \quad \chi\sim 1 \tag{C.8}$$

Then, the Sharpe Ratio (C.6) is twice as much as its initial value: $Sp(R)\sim 2\,Sp_0(R)$.

2. If the coefficient of correlation $a<0$ (C.3), the Sharpe Ratio $Sp(R)$ may decrease with increasing $\chi(W)$. For example, if a, $Sp_0(R)$, and χ are equal (C.9):

$$a\sim -\frac{1}{4} \quad ; \quad Sp_0^2(R) = \sim 2 \; ; \quad \chi\sim 1 \tag{C.9}$$

Then, the Sharpe Ratio (C.6) will be more than twice less than $Sp_0(R)$: $Sp(R)\sim 0{,}4\,Sp_0(R)$.

3. If the coefficient of correlation $a=0$ (C.3) is zero, the Sharpe Ratio $Sp(R)$ equals:

$$Sp(R) = \left[\frac{1+\chi^2}{\psi^2+\chi^2}\right]^{\frac{1}{2}}; \quad Sp(R)|_{\chi=0} = \psi_0^{-1} \quad ; \quad Sp(R)|_{\chi=\infty} = 1 \tag{C.10}$$

Appendix D. Price-Volume Correlations

Numerous empirical studies (Tauchen and Pitts, 1983; Karpoff, 1987; Campbell et al., 1993; DeFusco et al., 2017) present the "empirical evidence" of the positive or negative price-volume correlations. The general drawback of these empirical results is the implicit use of the mean price $\pi_j(t)$ (B.19), which correctly models mean price only when trade volumes are assumed constant. Such an assumption is completely wrong while estimating correlations between random trade volumes and prices. As it follows from (A.4-A.6), random trade volumes $U(t_i)$ define VWAP mean price $p(t)$ (A.6; D.1):

$$mE[p_j(t_i)] = p_j(t) = \frac{C_{\Sigma j}(t)}{U_{\Sigma j}(t)} = \frac{1}{U_{\Sigma j}(t)}\sum_{i=1}^{N} p_j(t_i)U_j(t_i) \tag{D.1}$$

The use of VWAP $p_j(t)$ (D.1), (A.4-A.6), and (B.5-B.7) gives:

$$C_j(t) = E[C_j(t_i)] = \frac{1}{N}\sum_{i=1}^{N} C_j(t_i) = E\left[p_j(t_i)U_j(t_i)\right] = \frac{1}{N}\sum_{i=1}^{N} p_j(t_i)U_j(t_i) = p_j(t)U_j(t) \tag{D.2}$$

The joint mathematical expectation of prices and volumes $E[p_j(t_i)U_j(t_i)]$ (D.2) equals the product of price-volume means (D.1; A.5):

$$E[p_j(t_i)U_j(t_i)] = p_j(t) \cdot U_j(t) = E[p_j(t_i)] \cdot E[U_j(t_i)] \tag{D.3}$$

The definition of correlation $corr\{p_j(t_i),U_j(t_i)\}$ between time series of prices $p_j(t_i)$ and trade volumes $U_j(t_i)$ (Shiryaev, 1999; Shreve, 2004) has the form (D.3):

$$corr\{p_j(t_i), U_j(t_i)\} = E[p_j(t_i)U_j(t_i)] - E[p_j(t_i)] \cdot E[U_j(t_i)] \tag{D.4}$$

Thus, (D.3) proves that the correlation $corr\{p_j(t_i),U_j(t_i)\}$ (D.4) equals zero. Random time series of prices $p_j(t_i)$ and trade volumes $U_j(t_i)$ always have zero correlation.

The positive or negative "empirical" correlation $corr\{p_j(t_i),U_j(t_i)\}|_{em}$ (D.5) of price-volume is the cause of the improper definition of the mean price $E[p_j(t_i)]=\pi_j(t)$ (B.19) that equals VWAP $p_j(t)$ (D.1) if one assumes that all trade volumes are constant. "Empirical studies" use mean price $\pi_j(t)$ (B.19) and mean volume $E[U_j(t_i)]=U_j(t)$ (A.5) and calculate correlation as:

$$corr\{p_j(t_i), U_j(t_i)\}|_{em} = \frac{1}{N}\sum_{i=1}^{N}(p_j(t_i) - \pi_j(t))(U_j(t_i) - U_j(t)) = \frac{1}{N}\sum_{i=1}^{N} p_j(t_i)\, U_j(t_i) - \pi_j(t)U_j(t) \tag{D.5}$$

From (D.5) and (D.2), obtain:

$$\frac{1}{N}\sum_{i=1}^{N} p_j(t_i)\, U_j(t_i) = p_j(t)U_j(t) \to corr\{p_j(t_i), U_j(t_i)\}|_{em} = [p_j(t) - \pi_j(t)] \cdot U_j(t) \tag{D.6}$$

The equation (D.6) proves that "empirical" assessments of $corr\{p_j(t_i),U_j(t_i)\}|_{em}$ (D.5) in reality calculate the difference between VWAP $p_j(t)$ (A.6) and $\pi_j(t)$ (B.19).

$$E[p_j(t_i)]|_{U_j(t_i)-const} = p_j(t)|_{U_j(t_i)-const} = \frac{1}{U_{\Sigma j}(t)}\sum_{i=1}^{N} p_j(t_i)U_j(t_i)\,|_{U_j(t_i)-const} = \pi_j(t) \tag{D.7}$$

The use of the mean price $\pi_j(t)$ (B.19; D.7), which equals VWAP $p_j(t)$ (D.1) when all *trade volumes are assumed constant*, is inconsistent with the attempt to calculate correlations between *random trade volumes* $U_j(t_i)$ and prices $p_j(t_i)$.

Practitioners and portfolio managers who search for market-based statistical dependence between random prices and trade volumes should use empirical time series to calculate the correlation $corr\{p_j(t_i), U^2_j(t_i)\}$ (D.11) between random price and ***squares*** of trade volumes:

$$corr\{p_j(t_i), U_j^{\,2}(t_i)\} = E\left[p_j(t_i), U_j^{\,2}(t_i)\right] - E\left[p_j(t_i)\right] \cdot E_m\left[U_j^{\,2}(t_i)\right] \qquad \text{(D.8)}$$

$$E\left[p_j(t_i), U_j^{\,2}(t_i)\right] = E\left[C_j(t_i) \cdot U_j(t_i)\right] = \frac{1}{N}\sum_{i=1}^{N} p_j(t_i) \cdot U_j^{\,2}(t_i) \qquad \text{(D.9)}$$

$$E\left[p_j(t_i)\right] = p_j(t), (\text{D.1}) \quad ; \quad E\left[U_j^{\,2}(t_i)\right] = \frac{1}{N}\sum_{i=1}^{N} U_j^{\,2}(t_i) \qquad \text{(D.10)}$$

$$corr\{p_j(t_i), U_j^{\,2}(t_i)\} = \frac{1}{N}\sum_{i=1}^{N} p_j(t_i) \cdot U_j^{\,2}(t_i) - p_j(t) \cdot \frac{1}{N}\sum_{i=1}^{N} U_j^{\,2}(t_i) \qquad \text{(D.11)}$$

The relations (D.8-D.11) are evident consequences of the Apps. A, and B.

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